Agentic AI Personal Loan Assistant

High-Level Design (HLD) & Infrastructure Architecture Specification

Document Version	v1.0.0 (Enterprise Production)	**Target Cloud**	Microsoft Azure
Core Stack	LangChain, LangGraph, FastAPI, Azure Functions	**Orchestration**	Azure Kubernetes Service (AKS)
Protocol	Agent-to-Agent (A2A) Standard Protocol	**Data Layer**	Azure Cosmos DB & Azure AI Search

1. ARCHITECTURE OVERVIEW

The **Agentic AI Personal Loan Assistant** is designed as a distributed multi-agent ecosystem capable of autonomously managing personal loan workflows—from pre-qualification and document verification to credit underwriting, risk assessment, and offer generation.

The platform leverages **LangGraph** to define deterministic state machines for complex multi-step reasoning, wrapped within **Python FastAPI** services for high-throughput REST and WebSocket interfaces. Inter-agent communication is governed by the standard **Agent-to-Agent (A2A) protocol**, ensuring standardized schema passing, agent discovery, and secure task delegation. Serverless **Azure Functions** execute event-driven asynchronous processing (e.g., OCR extraction, webhooks), while the entire core system is deployed on **Azure Kubernetes Service (AKS)** with enterprise-grade security and auto-scaling.

2. C4 MODEL — SYSTEM CONTEXT DIAGRAM (LEVEL 1)

The C4 Context diagram illustrates how the Agentic AI Personal Loan Assistant sits within the enterprise banking landscape, showing interaction boundaries between human users, internal processing units, external credit bureaus, and banking core services.

PERSON: BORROWER / APPLICANT

Applies for a personal loan, submits financial documents, and receives real-time agent updates.

PERSON: LOAN UNDERWRITER / OFFICER

Reviews flagged high-risk applications, inspects audit logs, and approves human-in-the-loop decisions.

↓ Interacts via Mobile/Web Portals (HTTPS / WebSockets)

SOFTWARE SYSTEM: AGENTIC AI PERSONAL LOAN ASSISTANT SYSTEM

Automates end-to-end loan application intake, document OCR parsing, policy RAG evaluation, risk assessment, and offer generation using A2A Protocol, FastAPI, LangChain/LangGraph on Azure (AKS & Functions).

Core Agents Breakdown:

3. LAYERED ARCHITECTURE DIAGRAM

The system is organized into a clean 6-layer architecture to decouple concerns across client interfaces, routing, multi-agent orchestration, async compute, storage, and infrastructure:

PRESENTATION LAYER	Web Portal (React/Next.js) • Mobile Native App • Embedded Partner Widgets (REST / WebSockets)
↓ **API Gateway Ingress**	
API & A2A PROTOCOL	Azure API Management (APIM) • OAuth2/OIDC • A2A Protocol Router • Rate Limiting & WAF
↓ **Agent Task Delegation**	
ORCHESTRATION LAYER	FastAPI App • LangGraph State Machine (Intake, Verification, Underwriting, Risk) • Human-in-the-Loop
↓ **Event Triggers / External Tool Calls**	
ASYNC WORKER LAYER	Azure Functions • Azure AI Document Intelligence (OCR) • Async Bank Statement Parsing
↓ **Persistence & Embedding Indexing**	
DATA & VECTOR LAYER	Azure Cosmos DB (State & Audit) • Azure AI Search (Vector RAG) • Azure Key Vault (Secrets)
↓ **Cloud Native Infrastructure**	
INFRASTRUCTURE LAYER	Azure Kubernetes Service (AKS) • Azure CNI • KEDA Auto-scaler • Azure OpenAI Service (GPT-4o)

4. AGENT-TO-AGENT (A2A) PROTOCOL SPECIFICATION